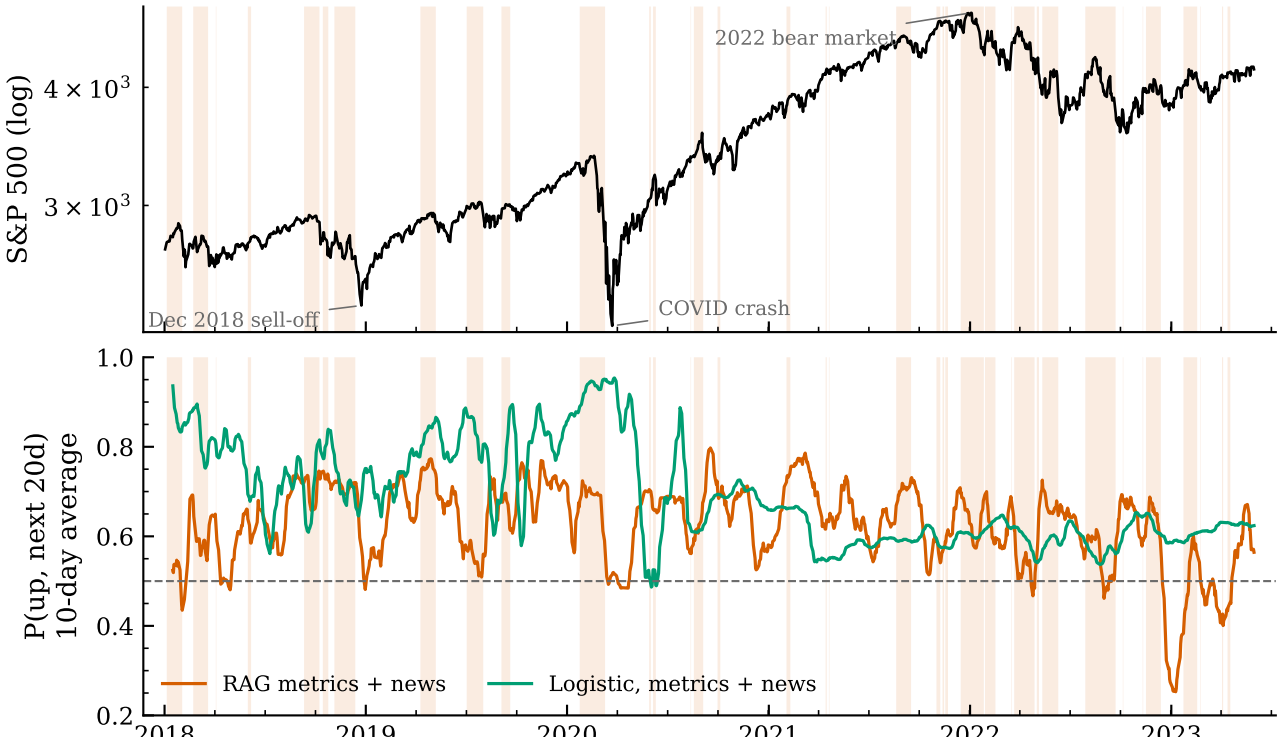

Held-out test: shaded = the next 20 trading days ended lower
S&P 500 (log)
4×10^3
3×10^3
2022 bear market
Dec 2018 sell-off
COVID crash
P(up, next 20d)
10-day average
1.0
0.8
0.6
0.4
0.2
RAG metrics + news
Logistic, metrics + news
2018
2019
2020
2021
2022
2023